

AI Interview Simulator

Comprehensive Project Architecture, Technical Specifications & Complete Details

Candidate: Srinivas Kandagatla | **Batch:** PFS-HYD-063 | **Stack:** React + Vite + TypeScript + FastAPI + Neon PostgreSQL + Google Gemini AI | **Status:** Production Live

1. Executive Summary & Project Purpose

The **AI Interview Simulator** is an advanced full-stack AI platform engineered to simulate end-to-end technical, behavioral, and coding interviews under authentic corporate hiring standards. It integrates real-time voice-to-voice conversational AI, dynamic resume-based contextual questioning, a live coding arena with **390 authentic LeetCode and GeeksForGeeks problems**, rigorous proctoring and anti-cheating mechanisms, comprehensive candidate analytics with downloadable PDF reports, and a multi-model failover pipeline powered by Google Gemini (gemini-2.5-flash & gemini-3.5-flash-lite).

Core Platform Pillar	Key Technical Implementation & Capabilities
Voice & Video AI Cockpit	Full-duplex voice interaction via Web Speech API (continuous Speech-to-Text) and natural Text-to-Speech synthesis. WebRTC video lobby with camera & mic diagnostics and automated hardware release.
Dynamic Adaptive AI Engine	Powered by Google Gemini 2.5 Flash with JSON schema enforcement. Formulates contextual questions based on candidate resume projects, previous answers, role level, and generates deep follow-ups.
Competitive Coding Arena	Monaco Editor integration with Python 3 & JavaScript execution. 390 curated DSA problems across 20+ algorithmic patterns with automated test case validation, parameter unpacking, and runtime metrics.
Enterprise Anti-Cheating	Full-screen lock enforcement, tab-switching visibility detection with 3-warning auto-termination, and a 90-second inactivity countdown modal.
Analytics & Scorecard Engine	Multi-dimensional candidate evaluation (Technical, Communication, Relevance, Completeness), filler word frequency analysis, mistake timeline, and dynamic ReportLab PDF scorecard generation.
Cloud Infrastructure & DevOps	FastAPI backend deployed on Render with non-blocking background DB initialization, React frontend on Vercel, Neon Serverless PostgreSQL, and automated 24/7 keep-alive pinging via Google Apps Script.

2. Complete Technology Stack & Tools Used

Layer / Domain	Technologies & Libraries	Specific Role & Justification
Frontend Framework	React 18, TypeScript, Vite 5	Modern component-based UI with strict type safety and sub-second hot module replacement.
Styling & Icons	Tailwind CSS 3.4, Lucide React	Utility-first dark/light responsive interface, glassmorphism accents, and crisp SVG iconography.
In-Browser Code IDE	@monaco-editor/react (VS Code Engine)	Full-featured code editor with syntax highlighting, indentation, auto-closing brackets, and themes.
Charts & Visuals	Recharts 2.12	Interactive performance radar charts, category breakdown bars, and historical progress lines.
Client Speech & Media	Web Speech API, WebRTC MediaDevices API	Native browser SpeechRecognition for speech-to-text, SpeechSynthesis for voice prompts, webcam stream.
Backend Web API	FastAPI 0.110, Python 3.12, Uvicorn 0.28	High-throughput asynchronous ASGI web framework with OpenAPI/Swagger interactive docs and Pydantic v2 schemas.
Database & ORM	Neon Serverless PostgreSQL, SQLAlchemy 2.0, Psycopg2-Binary	Cloud relational database with automated connection pooling, rollback safety, and ACID transaction guarantees.
Authentication & Security	Python-JOSE (JWT), Passlib (Bcrypt), OAuth2 Password Bearer	Cryptographically signed stateless JWT access tokens, salt-hashed passwords, role-based route guards.
Artificial Intelligence	Google Gemini API (gemini-2.5-flash & gemini-3.5-flash-lite)	Adaptive interview question generation, resume ATS parsing, multi-rubric answer scoring with strict JSON mode.
Code Execution Sandbox	Judge0 REST API & Isolated Subprocess Runner	Sandboxed execution for Python 3 and JavaScript with time-limit, memory constraints, and stdout/stderr capture.
PDF Report Engine	ReportLab 4.1, PyPDF 4.1	Publication-grade programmatic generation of candidate assessment scorecards with custom canvas page counters.
Cloud Hosting & DevOps	Render (Backend), Vercel (Frontend), Google Apps Script	Continuous deployment from GitHub; 5-minute HTTP GET keepalive script preventing Render free-tier sleep.

3. Deep-Dive Functional Modules

A. AI Live Mock Interview Cockpit

- **Hardware Diagnostics Lobby:** Interactive pre-flight check testing microphone audio level meters and live webcam mirror.
- **Full-Duplex Voice Dialogue:** The AI speaks questions aloud using natural web speech synthesis. The candidate replies using continuous voice recognition.
- **Contextual Adaptive Questioning:** Question #1 requests a personalized introduction; Question #2 conducts an architectural deep-dive into the candidate's actual resume projects; subsequent questions dynamically probe technical gaps identified from earlier answers.
- **Safe Device Cleanup:** Automatically terminates the MediaStream tracks, turning off camera LEDs and releasing audio recording immediately when the session ends.

B. Competitive Coding Arena (390 Curated DSA Problems)

- **20+ Algorithmic Patterns:** Comprehensive coverage of Two Pointers, Sliding Window, Monotonic Stack, Fast & Slow Pointers, BFS/DFS, Binary Search, Dynamic Programming, Heaps, Backtracking, and Greedy.
- **Authentic LeetCode & GFG Data:** Real test cases, problem statements, constraints, and curated Python 3 & JavaScript starter templates.
- **Smart Parameter Unpacking:** Automatically parses standard string test-case inputs and injects them into user solution functions seamlessly.

C. AI Resume Analyzer & Matcher

- **Deep ATS Parsing:** Extracts skills, projects, educational credentials, and work history from uploaded PDF resumes.
- **Target Role Fit Scoring:** Computes match percentages (0-100%) against specific job descriptions and identifies missing keywords.
- **Tailored Question Generation:** Automatically creates custom interview questions targeting specific technologies and claims in the resume.

D. Comprehensive Practice Question Bank

- **Categorized Question Catalog:** Hundreds of technical questions spanning DSA, System Design, SQL & DBMS, Python, Web Development, and Behavioral/HR.
- **Instant AI Answer Evaluation:** Gemini evaluates candidate answers with multi-rubric scoring (1-10), pinpointing exact strengths, omissions, and model 10/10 answers.

E. Anti-Cheating & Proctoring Integrity

- **Fullscreen Lock:** Enforces full-screen mode upon entering live interview sessions.
- **Tab Switching Violations:** Detects document visibility blur events. Displays two visible warning banners; the third violation immediately triggers automatic session submission.
- **Inactivity Guard:** Detects lack of candidate input for > 90 seconds, launching a 30-second warning countdown before auto-submitting.

4. Relational Database Design (Neon PostgreSQL)

Table Name	Key Schema Columns	Relationship & Description
users	id (PK), email, hashed_password, full_name, role, created_at	User accounts, authentication credentials, and system roles.
interviews	id (PK), user_id (FK), target_role, difficulty, status, total_questions, score	Interview session state machine: lobby, in_progress, completed, auto_submitted.
questions	id (PK), interview_id (FK), question_text, category, order_index, difficulty	Ordered sequence of questions generated or pulled for an interview.
answers	id (PK), question_id (FK), candidate_answer, score, feedback, duration_seconds	Speech-transcribed answers, latency, filler count, and AI scoring metrics.
interview_reports	id (PK), interview_id (FK), overall_score, technical_score, strengths, weaknesses	Persisted analytical assessment reports and mistake timelines.
coding_problems	id (PK), title, category, pattern, difficulty, examples, test_cases, templates	390 authentic LeetCode and GFG coding problems with solutions and test cases.
coding_submissions	id (PK), user_id (FK), problem_id (FK), code, language, status, runtime_ms	Candidate coding attempts, pass/fail status, execution time, and error logs.
practice_questions	id (PK), title, category, difficulty, question_text, sample_answer, explanation	Standardized question catalog for self-paced interview preparation.
practice_submissions	id (PK), user_id (FK), question_id (FK), user_answer, score, ai_feedback	Logged candidate attempts and scores on practice catalog items.
resumes	id (PK), user_id (FK), file_name, file_url, parsed_text, created_at	Uploaded candidate CVs, extracted raw text, and document metadata.
resume_analyses	id (PK), resume_id (FK), ats_score, skills, technologies, projects, suggestions	Structured AI analysis results, skill taxonomy, and project extractions.
proctoring_logs	id (PK), interview_id (FK), violation_type, timestamp, description	Audit trail of tab-switch events, fullscreen exits, and inactivity alerts.

5. Key REST API Endpoints Catalog

Endpoint Route	Method	Auth	Functionality & Responsibilities
/api/auth/register	POST	Public	User registration with email validation, bcrypt password hashing, and user creation.
/api/auth/login	POST	Public	Authenticates credentials and returns JWT bearer token and user role profile.
/api/interviews/start	POST	User	Creates interview session, parses resume context, and initializes Question #1.
/api/interviews/{id}/answer	POST	User	Submits candidate response, triggers Gemini evaluation, generates adaptive next question.
/api/interviews/{id}/finish	POST	User	Finalizes session, compiles comprehensive report, mistake timeline, and overall score.
/api/reports/{id}/pdf	GET	User	Dynamically streams high-resolution downloadable PDF scorecard built with ReportLab.
/api/coding/problems	GET	User	Lists 390 DSA coding problems filterable by pattern, difficulty, and category.
/api/coding/run	POST	User	Executes candidate code against sample test cases in sandboxed environment.
/api/coding/submit	POST	User	Evaluates all hidden test cases, records runtime/memory metrics and submission status.
/api/practice/questions/{id}/attempt	POST	User	AI scores candidate practice answer, returning strengths, weaknesses, and model answer.
/api/resume/analyze	POST	User	Uploads PDF resume, extracts skills, calculates ATS score, suggests interview questions.
/api/health & /	GET	Public	Liveness check and system metadata; target for keepalive monitoring scripts.

6. Cloud Deployment & Keep-Alive Architecture

- **Backend Deployment (Render):** Hosted as an ASGI web service running Uvicorn on Linux. Database table creation and initial seeding run inside a daemonized background thread so the server binds to the assigned port within 1.5 seconds, preventing Render startup deployment timeouts.
- **Database Tier (Neon PostgreSQL):** Serverless cloud PostgreSQL instance providing instant branching, connection pooling, and automated SSL encryption.
- **Frontend Deployment (Vercel):** Single-Page Application (SPA) deployed with automated Git CI/CD, route rewrites for clean client-side routing, and lightning-fast edge CDN distribution.
- **Zero-Sleep Keepalive (Google Apps Script):** A time-driven cron script triggers an HTTP GET request to the root API endpoint (/) every 5 minutes. This completely circumvents Render's 15-minute inactivity spin-down, ensuring 24/7 instant response times with zero cold-start latency.

7. Project Value & Competitive Advantages

The AI Interview Simulator sets a benchmark for modern AI-assisted career preparation. By unifying low-latency voice interaction, dynamic prompt intelligence, authentic competitive programming problem banks, and strict proctoring into a single cohesive platform, it provides candidates with an unparalleled, stress-tested simulation of genuine technical interviews.